

GRADUATE STUDY EDITION

# Hybrid-Electric Bus System Modeling

A MATLAB and Simulink textbook for backward-facing energy analysis,  
dual-battery supervisory control, route studies, and configuration optimization

Project database version 1.4.0

MATLAB / Simulink R2025a

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# Preface

**This book explains a complete concept-level hybrid-electric bus study environment. It is written for graduate students who already know basic mechanics, electrical power, and MATLAB syntax, but who may be new to model-based design and energy-management architecture.**

The project combines a transparent backward-facing physical model, an editable Simulink representation, an engineer-maintained Excel database, a programmatic MATLAB app, a bounded configuration optimizer, and automated verification. Its purpose is educational and architectural: it shows how equations, assumptions, data, software interfaces, and test evidence must remain consistent across a system model.

## How to read this book

Chapters 1-3 establish the system and data architecture. Chapters 4-10 derive the physical and supervisory models. Chapters 11-15 explain execution, optimization, routes, and validation. Chapters 16-17 interpret worked studies and define responsible extensions.

## Learning outcomes

- Derive tractive force and wheel power from a prescribed route speed and grade.
- Separate the traction DC-bus path from the genset-to-standby charging path.
- Explain a deterministic 30% dual-battery role-swap strategy and constant-point genset control.
- Distinguish energy accounting, equivalent replenishment, charge sustaining comparison, and true feasibility.
- Audit route provenance, component assumptions, numerical implementation, and validation evidence.
- Design extensions that increase fidelity without hiding uncertainty.

## Scope statement

This is a concept energy model. It does not claim certification, manufacturer performance, thermal safety, battery ageing, emissions compliance, or real-time controller readiness. Synthetic component values are used deliberately so that the architecture can be studied without implying supplier data.

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# Nomenclature and Conventions

**Table F.1. Principal symbols and units**

| Symbol / term         | Meaning                                                      | Typical unit          |
|-----------------------|--------------------------------------------------------------|-----------------------|
| $v, a$                | vehicle speed and longitudinal acceleration                  | m/s, m/s <sup>2</sup> |
| $m$                   | selected total vehicle mass                                  | kg                    |
| $F_{\text{trac}}$     | net tractive force required at the tyre contact patch        | N                     |
| $P_{\text{wheel}}$    | wheel mechanical power; positive traction, negative braking  | kW                    |
| $P_{\text{motor,dc}}$ | motor-side DC bus power; negative during regeneration        | kW                    |
| SOE                   | stored usable battery energy divided by usable-energy rating | fraction or %         |
| BSFC                  | brake-specific fuel consumption                              | g/kWh                 |
| $\eta$                | directional efficiency                                       | fraction              |
| $C_{\text{rr}}$       | rolling resistance coefficient                               | 1                     |
| $\rho$                | air density                                                  | kg/m <sup>3</sup>     |
| $C_d, A$              | drag coefficient and frontal area                            | 1, m <sup>2</sup>     |

## Sign convention

Battery power is positive when a pack discharges to the traction DC bus and negative while charging. Wheel power is positive for traction and negative for braking. Genset power is positive into the isolated standby charger; it never offsets traction demand.



# 1 Project Orientation

The project asks a systems question: for a prescribed bus mission, selected component set, loading state, environment, and supervisory calibration, how much wheel energy, battery energy, diesel fuel, grid-replenishment energy, operating cost, and feasible range result?

## 1.1 Why a backward-facing model?

A backward-facing model starts from the route speed that the vehicle is assumed to achieve. It differentiates speed to obtain acceleration, computes the tractive force required at the wheels, and then propagates power demand backward through the driveline to the DC bus and energy sources. This is efficient for energy sizing and comparison because it avoids solving a driver controller and vehicle-speed tracking loop.

### Interpretation boundary

A backward model estimates the energy and power that would be required to follow the trace. It does not prove that the selected vehicle can track the trace. The project therefore logs unmet traction energy whenever component limits prevent delivery.

## 1.2 Project deliverables

**Table 1.1. Primary project artifacts**

| Artifact                         | Role in the study                                                               |
|----------------------------------|---------------------------------------------------------------------------------|
| HybridBus_ComponentDatabase.xlsx | Master selections, catalogs, maps, routes, prices, and control calibrations     |
| HybridBus_BackwardModel.slx      | Editable ordinary-block Simulink representation with ten top-level subsystems   |
| simulate_hybrid_bus_core.m       | Detailed discrete reference kernel used by batch runs and optimization          |
| HybridBusApp.m                   | Explorer app for selecting, simulating, plotting, optimizing, and exporting     |
| tests/run_all_hybrid_bus_tests.m | Twenty-one assertion-based physics, isolation, control, cost, and ranking tests |

### 1.3 Architecture at a glance

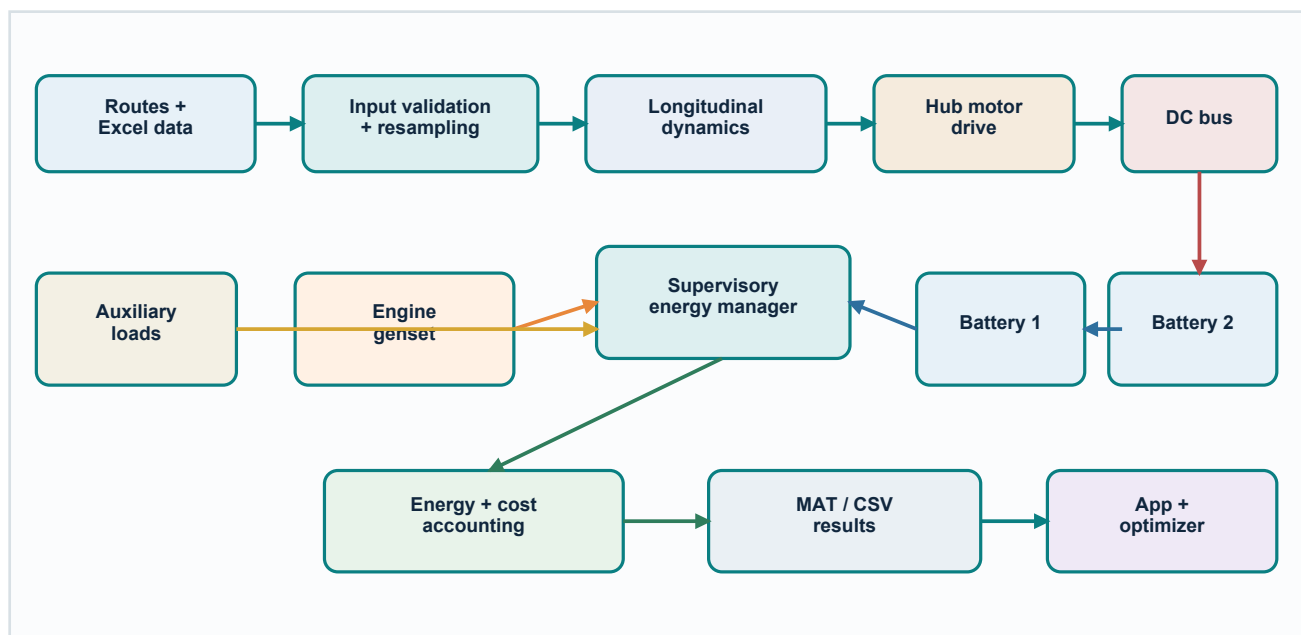


Figure 1.1. End-to-end project architecture from sourced routes to app and optimizer outputs

The Excel workbook is the single engineer-editable input source. The loader resolves stable IDs, the validator checks schema and physical ranges, and the input-preparation function resamples the selected route to a 1 s grid. Both the MATLAB kernel and the SLX model consume the same resolved parameter set.

## 2 Hybrid-Electric System Architecture

## 2.1 Energy topology

The architecture contains two independent battery packs, two rear hub motors, a diesel engine-generator set, and an auxiliary electrical load. The topology has two deliberately separated electrical paths. Only the active battery connects to the traction DC bus; the genset connects only to the standby battery through a dedicated charger. The genset therefore cannot propel the vehicle or reduce active-battery traction demand.

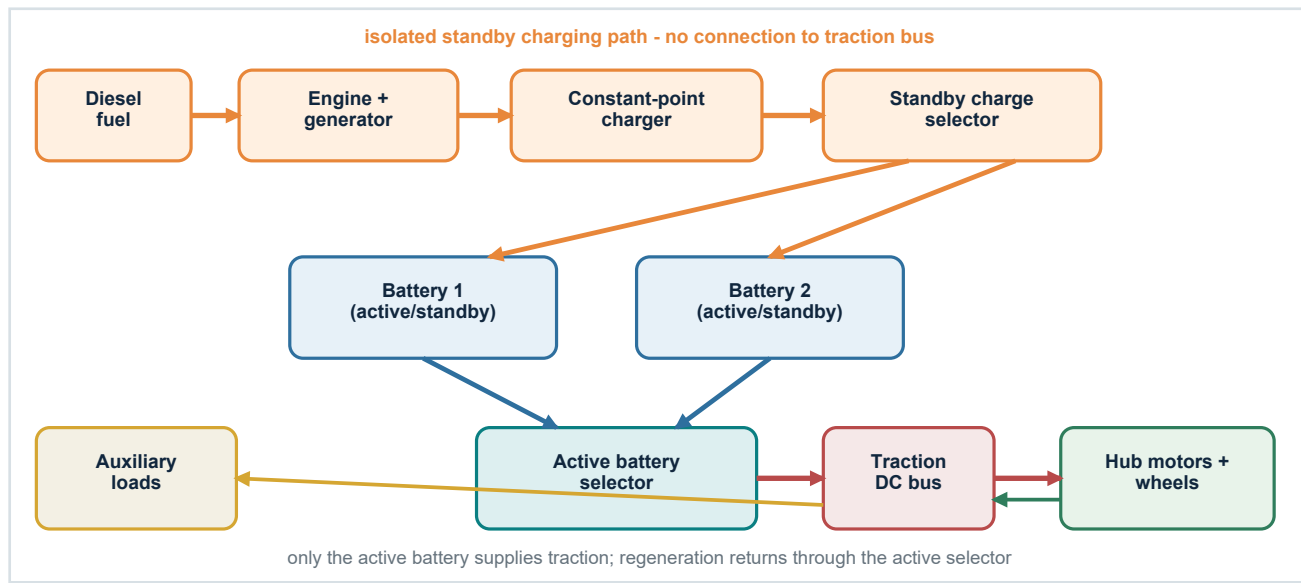


Figure 2.1. Conceptual energy flow and the project sign convention

## 2.2 Simulink decomposition

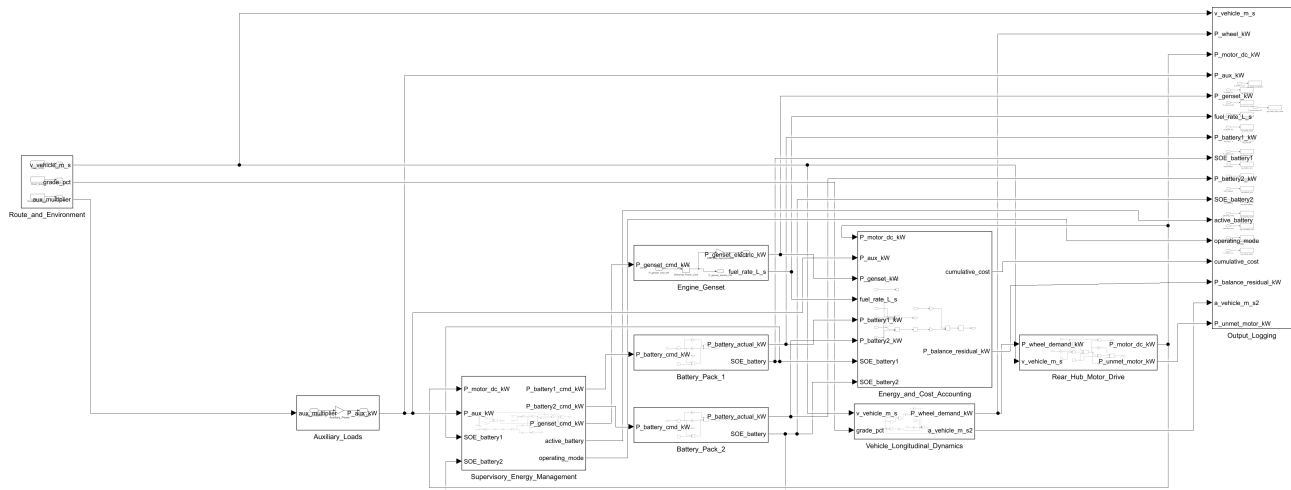


Figure 2.2. Top-level Simulink model and logged signal fan-out

**Table 2.1. Top-level model responsibilities**

| Subsystem                       | Responsibility                                                                       |
|---------------------------------|--------------------------------------------------------------------------------------|
| Route_and_Environment           | Publishes route speed, grade, and auxiliary multiplier                               |
| Vehicle_Longitudinal_Dynamics   | Computes filtered acceleration and required wheel power                              |
| Rear_Hub_Motor_Drive            | Applies motor torque, power, speed, efficiency, and regeneration limits              |
| Auxiliary_Loads                 | Calculates base and HVAC electrical demand                                           |
| Supervisory_Energy_Management   | Enforces the 30% role swap and routes fixed genset power only to standby             |
| Battery_Pack_1 / Battery_Pack_2 | Integrate independent energy states under power and SOE limits                       |
| Engine_Genset                   | Produces constant optimum electrical power when enabled and calculates fuel rate     |
| Energy_and_Cost_Accounting      | Checks the combined traction/charger boundary and accumulates energy/cost quantities |
| Output_Logging                  | Publishes stable workspace signals for inspection                                    |

## 2.3 Reference kernel versus SLX

The ordinary-block SLX is intended for architecture inspection, signal tracing, and interactive simulation. The MATLAB kernel contains the complete deterministic limit handling and supervisory behavior used by batch studies and the optimizer. This separation keeps the teaching model editable while making the numerical reference path easy to test.

## 3 Data Architecture and Reproducible Inputs

### 3.1 Workbook as engineering database

The version 1.4.0 workbook contains 23 sheets. IDs rather than row numbers form the interfaces between the Dashboard, catalogs, input resolver, app, and optimizer. This is important: a stable ID survives sorting and insertion, while a row number does not.

**Table 3.1. Workbook organization**

| Sheet group        | Examples                                                           | Purpose                                                     |
|--------------------|--------------------------------------------------------------------|-------------------------------------------------------------|
| Selections         | Dashboard                                                          | Chosen component IDs, SOEs, prices, fuel tank, and controls |
| Component catalogs | Battery, Motor, Genset, Tyre, Final Drive, Mass                    | Ratings, efficiency, compatibility, and enable flags        |
| Maps               | Engine_Fuel_Map, Generator_Efficiency_Map                          | Bounded lookup data for fuel and conversion efficiency      |
| Routes             | Route_Catalog, Route_Time_Speed, Route_Distance_Speed, Route_Grade | Provenance and mission traces                               |
| Calibration        | Environment, Control_Calibration, Energy_Prices                    | Scenario and supervisor parameters                          |
| Governance         | Units_and_Definitions, Change_Log                                  | Definitions and database history                            |

### 3.2 Input pipeline

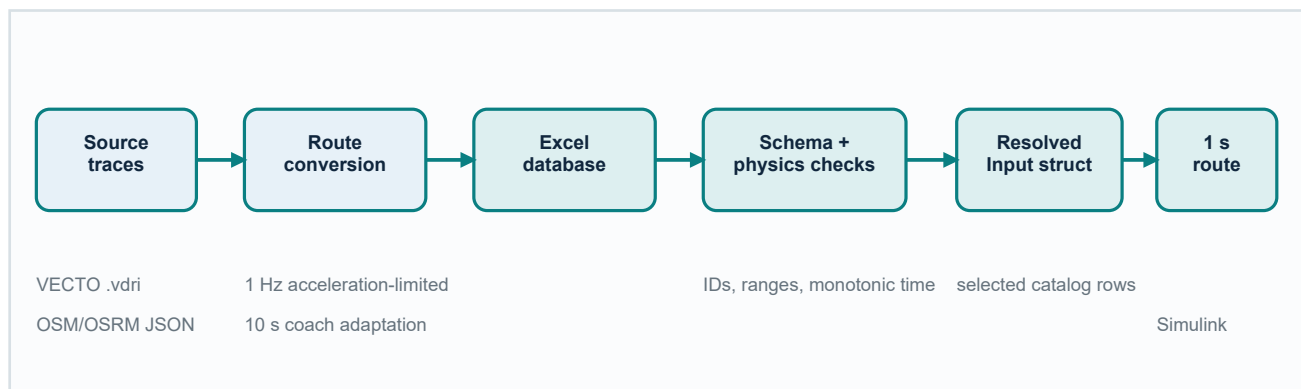


Figure 3.1. Route and component data pipeline

The loader imports every sheet with preserved column names. Validation then checks required sheets, unique IDs, battery and motor consistency, total-mass positivity, route finiteness, monotonic time, non-negative speed, acceleration bounds, and nonzero distance. Preparation resolves selected rows into scalar structures and resamples speed, grade, and auxiliary multiplier at the model sample time.

### 3.3 Default configuration

**Table 3.2. Default case resolved from the Dashboard**

| Input                    | Selection / value | Unit or interpretation                |
|--------------------------|-------------------|---------------------------------------|
| Route                    | VECTO-URBAN       | stable route ID                       |
| Battery 1 / 2            | BAT-12 / BAT-12   | independent packs                     |
| Motor / genset           | MOT-12 / GEN-05   | two identical hub motors + one genset |
| Total mass               | 19,000            | kg                                    |
| Initial B1 / B2 SOE      | 85 / 75           | %                                     |
| Fuel / electricity price | 1.65 / 0.18       | EUR/L / EUR/kWh                       |

## 4 Longitudinal Vehicle Dynamics

### 4.1 Speed, acceleration, and grade

The route speed is prescribed. A first-order discrete filter is applied to the raw finite-difference acceleration to reduce numerical spikes without introducing a continuous state.

$$a_{\text{raw},k} = (v_k - v_{k-1}) / \Delta t_{k-1} \quad (4.1)$$

$$a_k = a_{k-1} + [\Delta t / (\tau + \Delta t)] (a_{\text{raw},k} - a_{k-1}) \quad (4.2)$$

Road grade is stored as percent. The model converts it to an angle using  $\theta = \text{atan}(\text{grade}/100)$ . Headwind is added to vehicle speed and clipped so aerodynamic relative speed cannot become negative.

### 4.2 Road-load forces

$$F_{\text{inertia}} = m a \quad (4.3)$$

$$F_{\text{roll}} = m g C_{rr} \cos(\theta) \quad (4.4)$$

$$F_{\text{grade}} = m g \sin(\theta) \quad (4.5)$$

$$F_{\text{aero}} = 0.5 \rho C_d A v_{\text{air}}^2 \quad (4.6)$$

$$F_{\text{tractive}} = F_{\text{inertia}} + F_{\text{roll}} + F_{\text{grade}} + F_{\text{aero}} \quad (4.7)$$

$$P_{\text{wheel,demand}} = F_{\text{tractive}} v / 1000 \quad (4.8)$$

#### Mass definition

The selected TotalVehicleMass\_kg is the complete simulated mass. It already represents curb mass, passengers, cargo, batteries, motors, final drives, and genset. Component mass is reported separately for traceability and is not added a second time.

### 4.3 Force scaling intuition

Inertia, rolling resistance, and grade forces scale linearly with total mass. Aerodynamic force does not. Therefore mass strongly changes stop-and-go and climbing demand, while high-speed cruise also depends heavily on frontal area and drag coefficient.

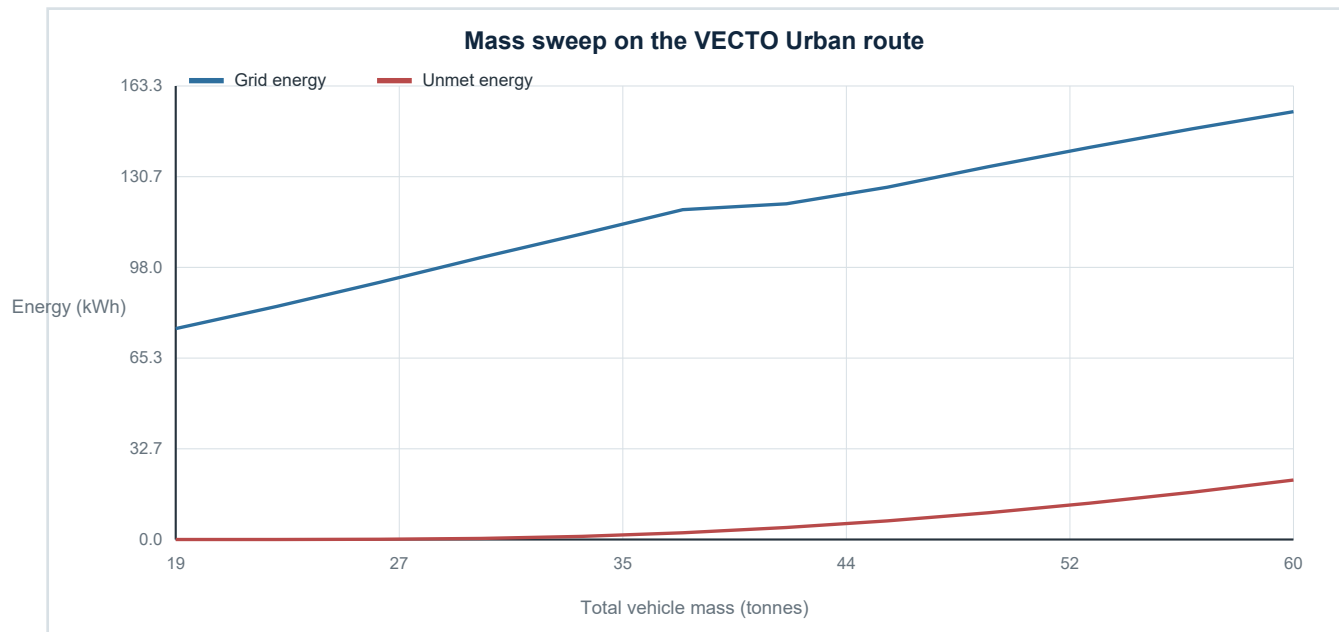


Figure 4.1. Energy and feasibility sensitivity from 19 to 60 tonnes

The default component set remains essentially feasible at the low end, but unmet traction energy grows with mass. Cost per kilometre alone must not be used to rank infeasible cases because unsupplied energy is not purchased.



## 5 Rear Hub Motors and Fixed Driveline

### 5.1 Kinematics

$$\omega_{\text{wheel}} = v / r_{\text{loaded}} \quad (5.1)$$

$$\omega_{\text{motor}} = i_{\text{fd}} \omega_{\text{wheel}}, \quad n_{\text{motor}} = 60 \omega_{\text{motor}} / (2\pi) \quad (5.2)$$

The model uses two identical rear hub motors. The selected fixed reduction ratio connects wheel speed to motor speed. If motor speed exceeds the catalog maximum, available motor power is set to zero and the shortfall is logged.

### 5.2 Torque and power envelope

$$T_{\text{available}} = \min[T_{\text{peak}}, P_{\text{peak}} / \max(\omega_{\text{motor}}, \epsilon)] \quad (5.3)$$

$$P_{\text{tractive,max}} = 2 \min(P_{\text{peak}}, T_{\text{available}} \omega_{\text{motor}}) \eta_{\text{fd,mot}} \quad (5.4)$$

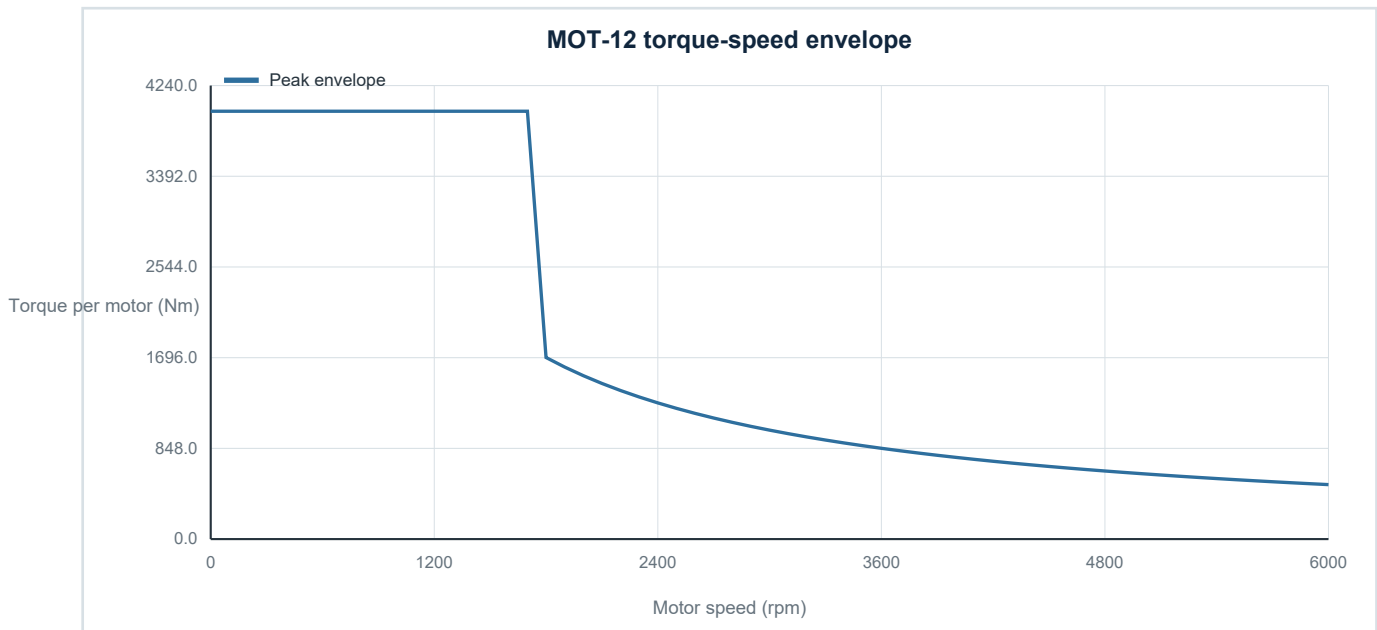


Figure 5.1. Selected motor constant-torque and constant-power regions

### 5.3 Directional efficiency and regeneration

Traction wheel power is divided by motor and final-drive motoring efficiencies to obtain DC demand. During braking, delivered negative wheel power is multiplied by regeneration efficiencies, producing negative motor DC power. Regeneration is limited by motor capability, fixed-drive direction, and the lower of the two pack regeneration ratings.

$$P_{\text{motor,dc}} = P_{\text{wheel}} / (\eta_{\text{fd,mot}} \eta_{\text{motor,mot}}) \text{ for traction} \quad (5.5)$$

$$P_{\text{motor,dc}} = P_{\text{wheel}} \eta_{\text{fd,reg}} \eta_{\text{motor,reg}} \text{ for regeneration} \quad (5.6)$$

## 6 Battery Energy Model

### 6.1 Energy state and SOE

Each battery is represented by usable stored energy rather than voltage, current, temperature, or electrochemical states. This is appropriate for concept energy analysis but cannot predict electrical transients or thermal derating.

$$\text{SOE} = E / E_{\text{usable}} \quad (6.1)$$

$$E_{k+1} = E_k - P_{\text{dis}} \Delta t / (3600 \eta_{\text{dis}}) \quad (6.2)$$

$$E_{k+1} = E_k + |P_{\text{chg}}| \eta_{\text{chg}} \Delta t / 3600 \quad (6.3)$$

### 6.2 Three simultaneous limits

At every time step the battery helper applies the minimum of the requested power, catalog power limit including derating, and energy-implied power needed to remain within minimum or maximum SOE. Charge power is additionally bounded by the pack regeneration rating.

#### Numerical safeguard

The energy state is clamped to [MinSOE, MaxSOE] after each update. Any source or load that cannot be accepted is logged as rejected charge or unmet DC power; it is not silently discarded.

### 6.3 Why one active traction pack?

The supervisor permits only one battery to discharge for traction. This is intentionally conservative: the active pack must be able to support residual DC demand by itself. The rule also creates a clean architecture for switching, standby charging, and deterministic regenerative routing.

**Table 6.1. Synthetic battery catalog coverage**

| Catalog range     | Minimum | Maximum |
|-------------------|---------|---------|
| Usable energy     | 100 kWh | 200 kWh |
| Maximum discharge | 300 kW  | 500 kW  |
| Maximum charge    | 150 kW  | 280 kW  |
| Mass              | 850 kg  | 1450 kg |

## 7 Engine-Generator Set and Fuel Model

### 7.1 Constant best-efficiency operating point

The genset has one permitted electrical command while on: the catalog OptimumPower\_kW value, bounded by the rated maximum. It is not load-following and is not connected to the traction bus. Its dedicated charger sends power only to the pack currently designated standby. Battery charge-power and upper-SOE limits determine accepted power; any excess is recorded as rejected charge.

### 7.2 Generator and engine maps

$$P_{\text{mech}} = P_{\text{gen,dc}} / \eta_{\text{gen}}(\lambda) \quad (7.1)$$

$$\dot{m}_{\text{fuel}} = \max[\dot{m}_{\text{idle}}, P_{\text{mech}} \text{BSFC}(\lambda) / (1000 \rho_{\text{fuel}} 3600)] \quad (7.2)$$

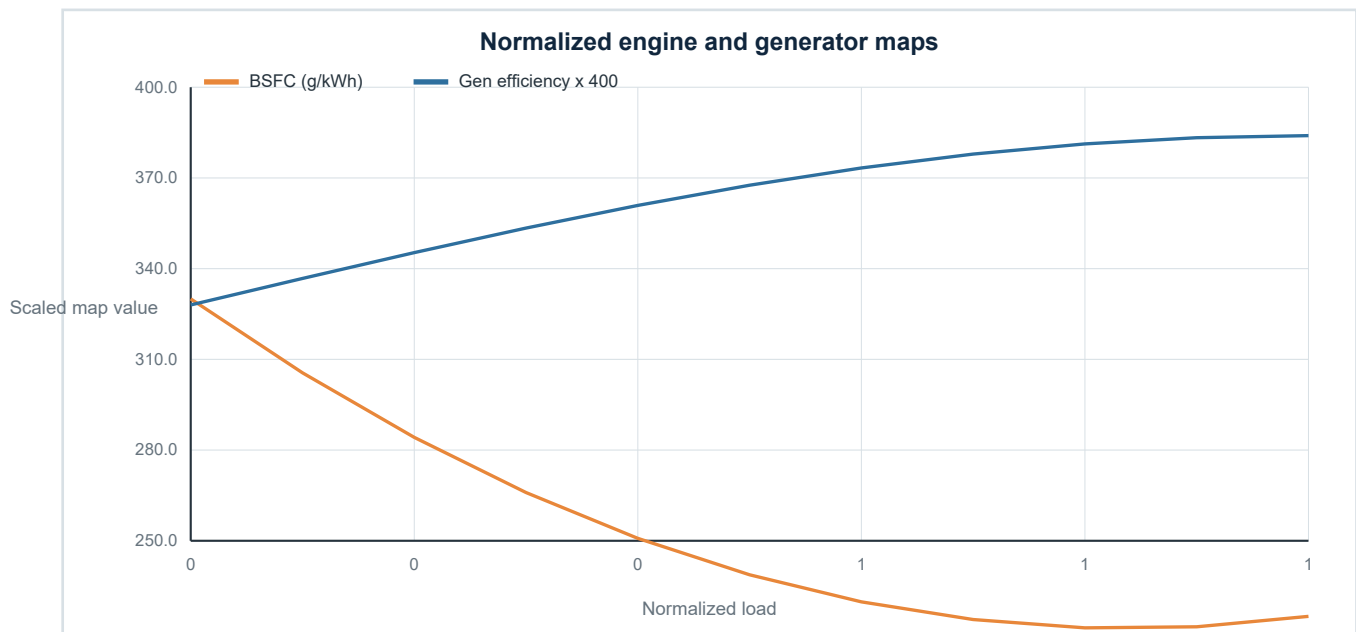


Figure 7.1. Synthetic BSFC and generator-efficiency maps used for interpolation

The generator efficiency is clipped to a physically reasonable numerical range, and BSFC is linearly interpolated over normalized load. A fixed start-fuel penalty is added on every off-to-on transition.

### 7.3 What the fuel model does not claim

#### Not an emissions model

There is no engine warm-up, transient combustion, aftertreatment, ambient correction, fuel-temperature effect, or pollutant calculation. The output is a concept diesel-volume estimate, not an emissions or homologation result.

# 8 Supervisory Energy Management

## 8.1 Control objectives

The deterministic supervisor has four priorities: keep the genset electrically isolated from traction; preserve exactly one active traction pack; swap roles at 30% SOE when the alternate pack is ready; and recharge the depleted standby pack at constant best-efficiency genset power while respecting battery limits.

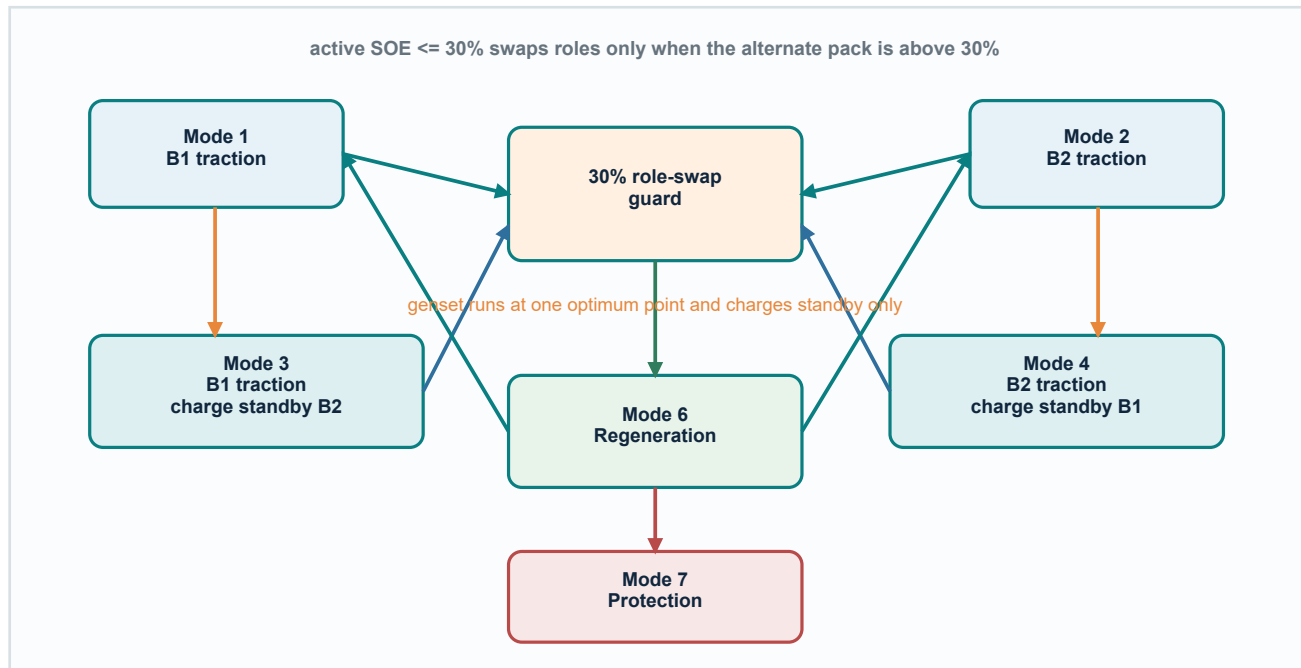


Figure 8.1. Supervisory operating modes and principal transitions

## 8.2 Decision order

- 1 Read both pack SOEs and retain the current active-battery state.
- 2 If active SOE is at or below 30% and the standby pack is above 30%, exchange the active and standby roles immediately.
- 3 Start the genset when standby SOE is at or below 30%; while on, command the constant OptimumPower\_kW value.
- 4 Send positive motor-plus-auxiliary demand only to the active battery; the genset is excluded from this equation.
- 5 Send genset power only to the standby battery. Allocate regeneration independently to an eligible battery.
- 6 Update energy states, fuel, operating mode, losses, rejected charge, and the balance residual.

## 8.3 Hysteresis and dwell time

The 30% role threshold is an architectural rule rather than a tunable calibration. A role swap is allowed only if the alternate pack is above 30%; this prevents rapid swapping when both packs are depleted. If both are low, the current active pack remains connected while the genset charges standby. Charging continues until the standby pack reaches its upper SOE limit or becomes the active pack.

**Table 8.1. Operating-mode interpretation**

| Mode | Meaning               | Primary power behavior                                   |
|------|-----------------------|----------------------------------------------------------|
| 1    | Battery 1 active      | B1 alone supplies positive traction-bus demand           |
| 2    | Battery 2 active      | B2 alone supplies positive traction-bus demand           |
| 3    | B1 active; B2 charged | Genset fixed output is routed only to standby B2         |
| 4    | B2 active; B1 charged | Genset fixed output is routed only to standby B1         |
| 6    | Regeneration          | Negative motor DC power is allocated to an eligible pack |
| 7    | Protection            | Sources are clamped and unmet demand is recorded         |

# 9 Power, Energy, Cost, and Range Accounting

## 9.1 DC power conservation

Two node equations make the isolation explicit. At the traction node, only the active battery balances motor and auxiliary demand. At the charger node, the genset balances standby-battery charging and rejected power. Summing the nodes gives the project-wide residual without implying a physical genset-to-traction connection.

$$P_{\text{active}} - P_{\text{motor,dc}} - P_{\text{aux}} + P_{\text{unmet}} = 0 \quad (9.1a)$$

$$P_{\text{gen}} + P_{\text{standby}} - P_{\text{rejected}} = 0 \quad (9.1b)$$

$$P_{\text{gen}} + P_{B1} + P_{B2} - P_{\text{motor,dc}} - P_{\text{aux}} + P_{\text{unmet}} - P_{\text{rejected}} = P_{\text{residual}} \quad (9.1c)$$

## 9.2 Equivalent grid replenishment

$$E_{\text{grid}} = [\max(0, E_{1,0} - E_{1,f}) + \max(0, E_{2,0} - E_{2,f})] / \eta_{\text{grid}} \quad (9.2)$$

$$C_{\text{total}} = V_{\text{fuel}} c_{\text{fuel}} + E_{\text{grid}} c_{\text{electricity}} \quad (9.3)$$

$$C_{\text{km}} = C_{\text{total}} / d_{\text{route}} \quad (9.4)$$

Terminal battery surplus is not credited by default. This prevents an artificial negative electric cost and makes the equivalent-replenishment comparison conservative.

## 9.3 Range estimates

$$R_{\text{battery}} = E_{\text{usable,initial}} / e_{\text{DC,route}} \quad (9.5)$$

$$R_{\text{fuel}} = V_{\text{tank}} / v_{\text{fuel,per-km}} \quad (9.6)$$

### Range is conditional

Range estimates assume that the observed route energy intensity and supervisory behavior remain representative. If the example route never starts the genset, the code estimates supported fuel range at the selected optimum genset point rather than reporting infinity.

## 9.4 Feasibility before economics

A low cost per kilometre is meaningful only when unmet traction energy is negligible and the balance residual satisfies tolerance. An infeasible case can look artificially cheap because energy that the vehicle failed to deliver was never purchased. The optimizer therefore filters feasibility before ranking cost.

# 10 Numerical Implementation in MATLAB and Simulink

## 10.1 Discrete solution

The default solver interpretation is fixed-step discrete with a 1 s sample time. Route speed, grade, and auxiliary multiplier are linearly interpolated; stop flags use previous-value interpolation. Energy updates use time-step integration in kWh, while fuel rate is integrated in litres.

## 10.2 Execution sequence

```
cd('C:\TempData\Hybrid_Vehicle\HybridBusProject')
project = startup_hybrid_bus;
Results = run_hybrid_bus_simulation;
open_system('HybridBus_BackwardModel.slx')
app = HybridBusApp;
```

The runner loads and validates the database, resolves the selected input structure, executes the detailed kernel, attaches validation metadata, and optionally exports MAT and CSV files. The SLX workspace publisher assigns only the stable, unit-suffixed variables needed by the block diagram.

## 10.3 Results structure

**Table 10.1. Simulation output contract**

| Field                 | Contents                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------|
| Metadata              | database file/version, MATLAB release, model version, timestamp                             |
| SelectedConfiguration | stable IDs for route and every selected component/calibration                               |
| InputParameters       | fully resolved scalar structures, maps, and resampled route                                 |
| Signals               | grouped vehicle, wheel, motor, auxiliary, battery, genset, controller, and energy histories |
| Summary               | distance, fuel, grid energy, costs, range, SOE, unmet energy, residual, and mass            |
| Validation            | warnings, constraint violations, and feasibility flag                                       |

# 11 Using the Explorer App

## 11.1 Configuration panel

The programmatic UIFigure app presents route, battery, motor, genset, total-mass, and auxiliary selections in a fixed left sidebar. Initial battery SOEs are entered as percentages and converted to fractions only at the model boundary. Fuel and electricity price units are explicit.

The Powertrain Architecture tab shows the isolated standby-charging chain, active-battery selector, traction DC bus, regenerative path, and supervisory command links. It is the quickest visual check that genset power has no direct route to the motors or auxiliaries.

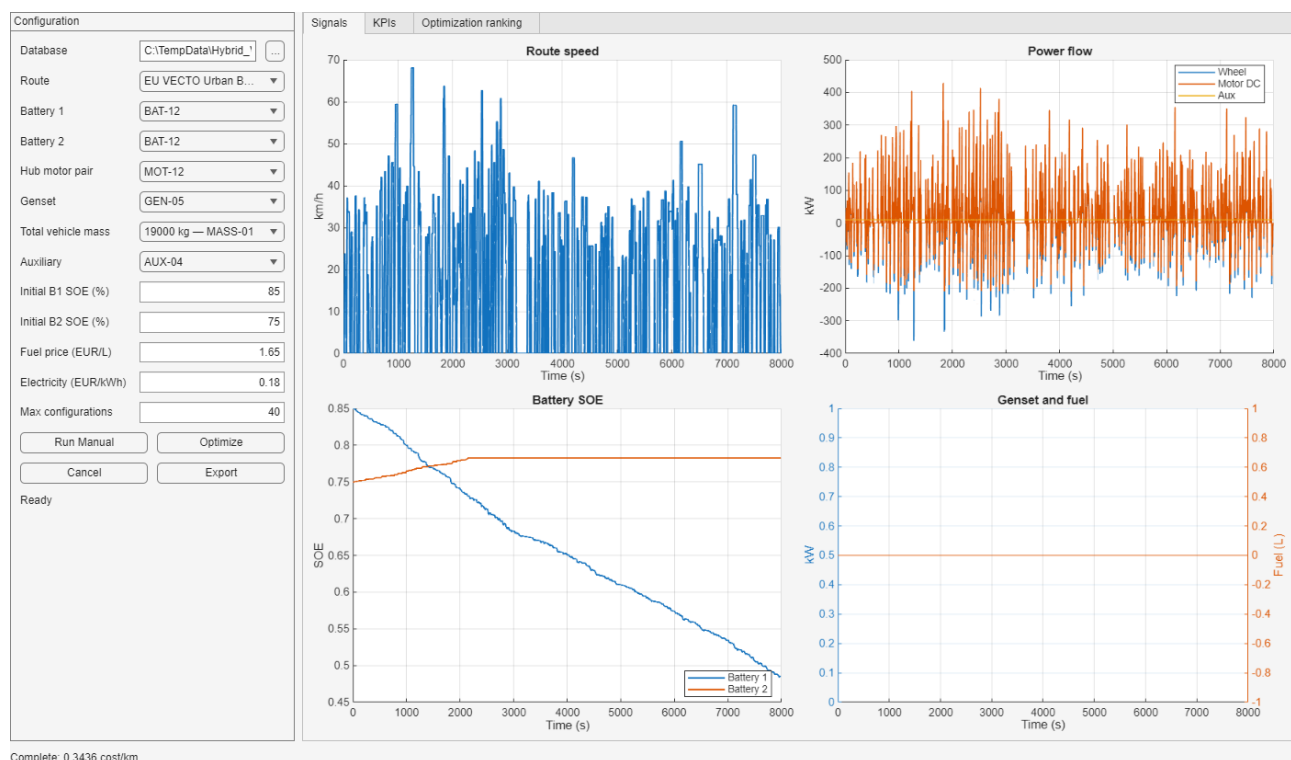


Figure 11.1. Explorer app after running the default VECTO Urban case

## 11.2 Manual-study workflow

- 1 Select the database and confirm that the status reports a validated version.
- 2 Choose a route, component set, total vehicle mass, initial SOEs, and prices.
- 3 Run the manual case and inspect speed, power flow, battery SOE, genset/fuel, KPIs, and warnings.
- 4 Treat any unmet traction energy as a failed design case, not as an acceptable low-cost solution.
- 5 Export the selected result only after checking assumptions and feasibility.

## 11.3 Optimization tab

The optimization view ranks feasible configurations and keeps stable IDs visible. Cancellation is checked between simulations, so a bounded search remains responsive without requiring Parallel Computing Toolbox.



# 12 Configuration Optimization

## 12.1 Bounded mixed-radix enumeration

Candidate IDs are assembled from enabled catalog rows. A mixed-radix counter converts one linear index into a component combination without generating a huge Cartesian table in memory. Search is bounded by the requested maximum and by a larger attempt limit that allows incompatible cases to be skipped.

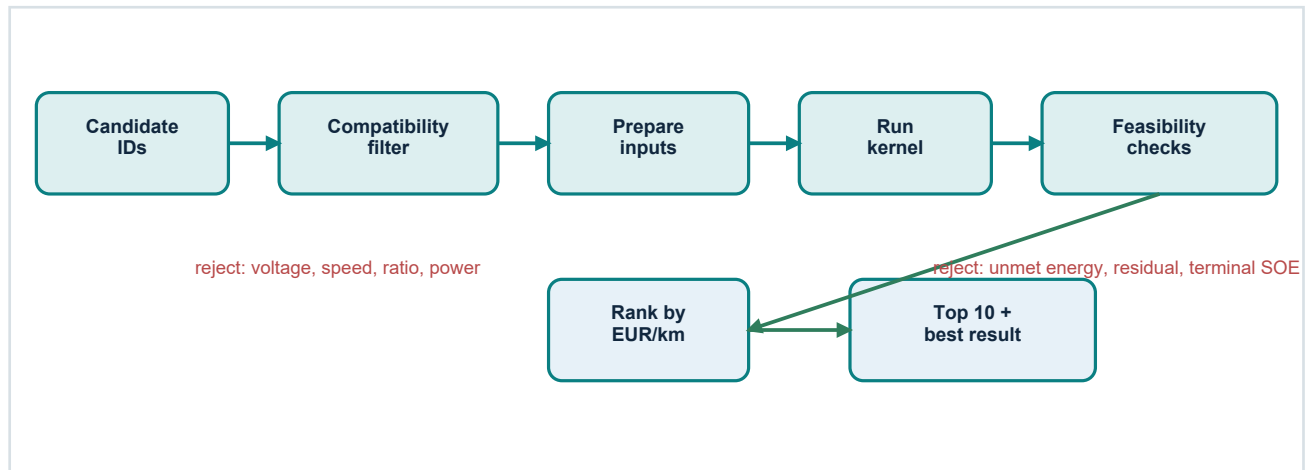


Figure 12.1. Compatibility, simulation, feasibility, and ranking sequence

## 12.2 Compatibility filters

- Battery-to-motor DC voltage class difference no greater than 50 V.
- Motor speed below maximum at the selected route speed, tyre radius, and final-drive ratio.
- Final-drive ratio inside the selected motor compatibility interval.
- At least one selected battery has adequate continuous discharge rating for the motor rating.
- Genset optimum power is within its rating and within either standby battery's charge-power capability.

## 12.3 Objective and feasibility

$$\text{minimize } C_{\text{km}} \text{ subject to } E_{\text{unmet}} < 10^{-3} \text{ kWh, } E_{\text{balance}} \leq \text{tolerance, and terminal-SOE compliance} \quad (12.1)$$

A current 144-case Battery 1 / Motor sweep produced 13 feasible configurations. The best feasible row used BAT-11 with MOT-12 at approximately 0.3429 EUR/km. The default BAT-12 / MOT-12 case ranked very close at approximately 0.3436 EUR/km. Because all component values are synthetic, this ranking demonstrates workflow rather than a procurement recommendation.

**Table 12.1. Top five feasible rows in the current 144-case study**

| Rank | Battery 1 | Motor  | Cost (EUR/km) | Electricity (kWh/km) |
|------|-----------|--------|---------------|----------------------|
| 1    | BAT-11    | MOT-12 | 0.3429        | 1.9050               |
| 2    | BAT-12    | MOT-12 | 0.3436        | 1.9091               |
| 3    | BAT-10    | MOT-12 | 0.3440        | 1.9110               |
| 4    | BAT-09    | MOT-12 | 0.3445        | 1.9137               |
| 5    | BAT-11    | MOT-11 | 0.3459        | 1.9216               |

# 13 Route Library and Provenance

## 13.1 Route categories

The database contains 20 routes: three official EU VECTO passenger missions, eight geographic long-distance coach corridors, and nine synthetic sensitivity profiles. The Route\_Catalog records the region, source basis, organization, URL, version or retrieval time, licence, distance, duration, maximum speed, and notes.

## 13.2 VECTO missions

The Urban, Suburban, and Coach declaration-mode mission cycles are unmodified source files from the European Commission VECTO repository. The converter applies a deterministic bus driver with 1.0 m/s<sup>2</sup> acceleration and 1.3 m/s<sup>2</sup> braking bounds, honors declared stops, preserves the raw distance-domain trace, and produces a 1 Hz time history for this model.

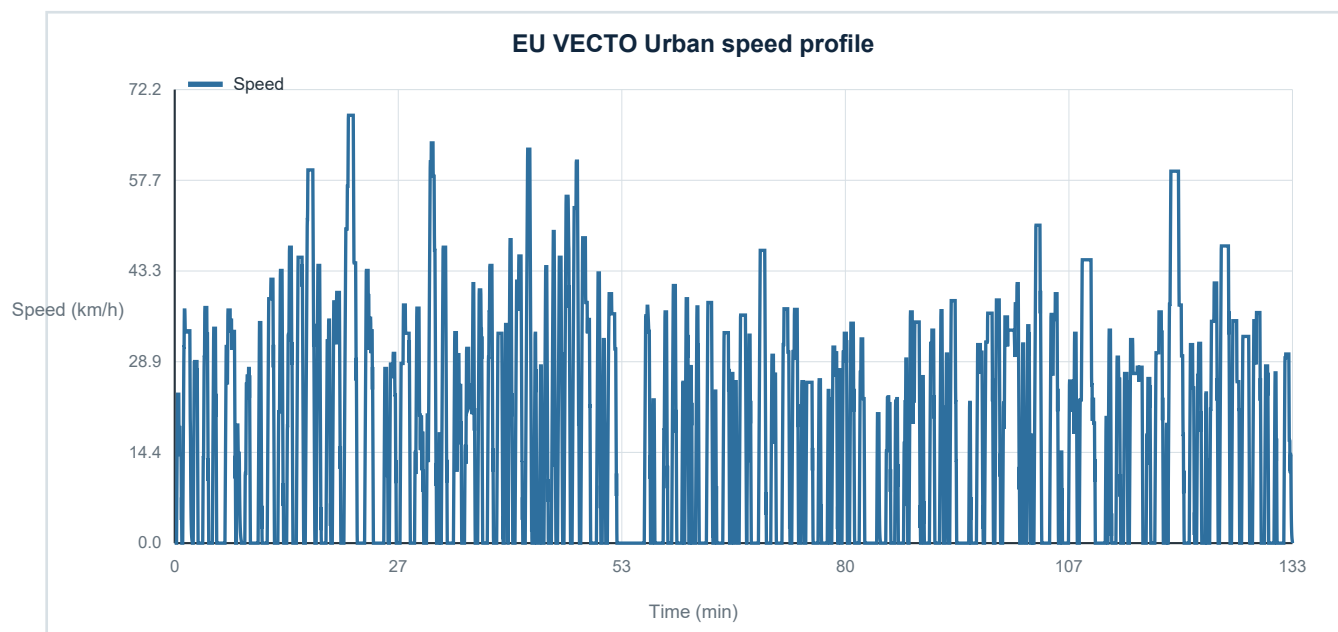


Figure 13.1. Default EU VECTO Urban mission after deterministic time conversion

### 13.3 Long European corridors

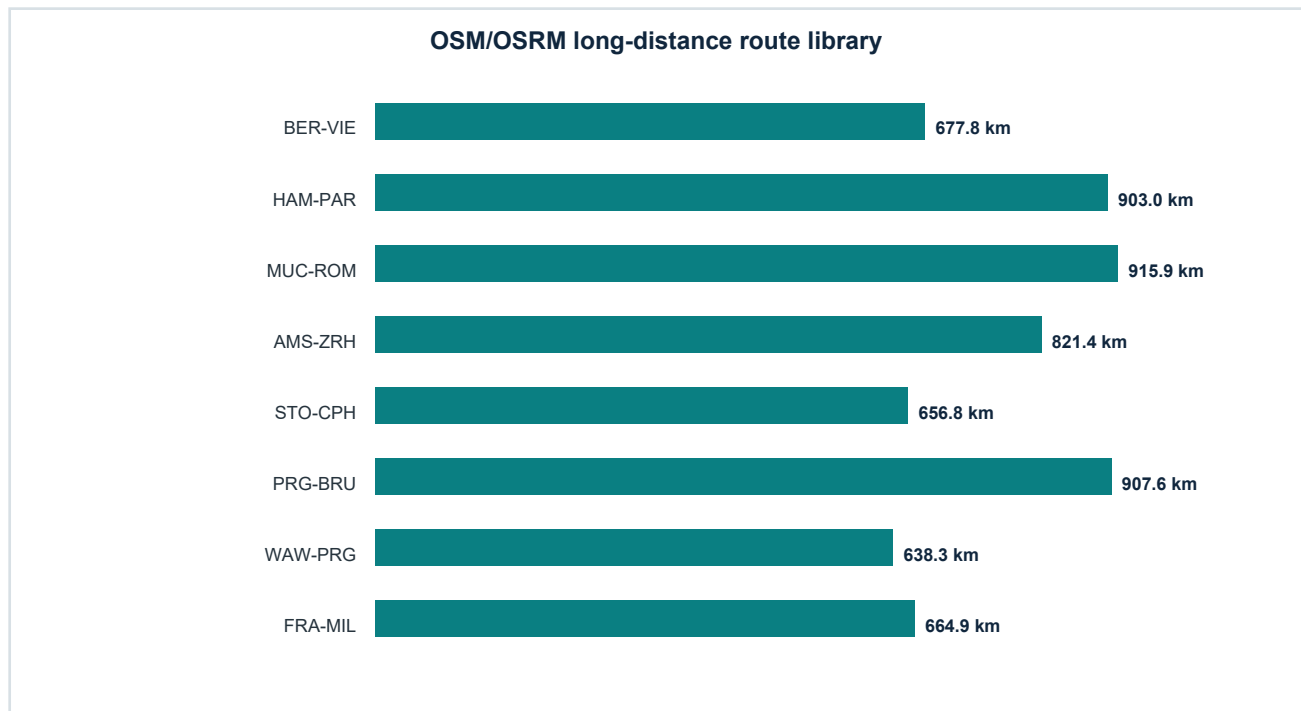


Figure 13.2. Geographic coach corridors in the requested 600-1000 km band

**Table 13.1. Long-route source distances and adapted mission durations**

| Route                                         | Countries                           | Distance | Adapted duration |
|-----------------------------------------------|-------------------------------------|----------|------------------|
| Berlin, Germany to Vienna, Austria            | Germany / Austria                   | 677.8 km | 8.45 h           |
| Hamburg, Germany to Paris, France             | Germany / France                    | 903.0 km | 11.18 h          |
| Munich, Germany to Rome, Italy                | Germany / Austria / Italy           | 915.9 km | 11.54 h          |
| Amsterdam, Netherlands to Zurich, Switzerland | Netherlands / Germany / Switzerland | 821.4 km | 9.73 h           |
| Stockholm, Sweden to Copenhagen, Denmark      | Sweden / Denmark                    | 656.8 km | 8.34 h           |
| Prague, Czechia to Brussels, Belgium          | Czechia / Germany / Belgium         | 907.6 km | 11.12 h          |
| Warsaw, Poland to Prague, Czechia             | Poland / Czechia                    | 638.3 km | 8.32 h           |
| Frankfurt, Germany to Milan, Italy            | Germany / Switzerland / Italy       | 664.9 km | 8.34 h           |

#### Route fidelity warning

OSRM supplies road-network distance and estimated segment durations but not elevation. Long-route speed is capped at 100 km/h, a 45-minute stationary break is inserted after each 4.5-hour driving block, and grade is set to zero. Mountain-route energy will therefore be understated until elevation is added.

# 14 Verification, Validation, and Test Evidence

## 14.1 Layered evidence

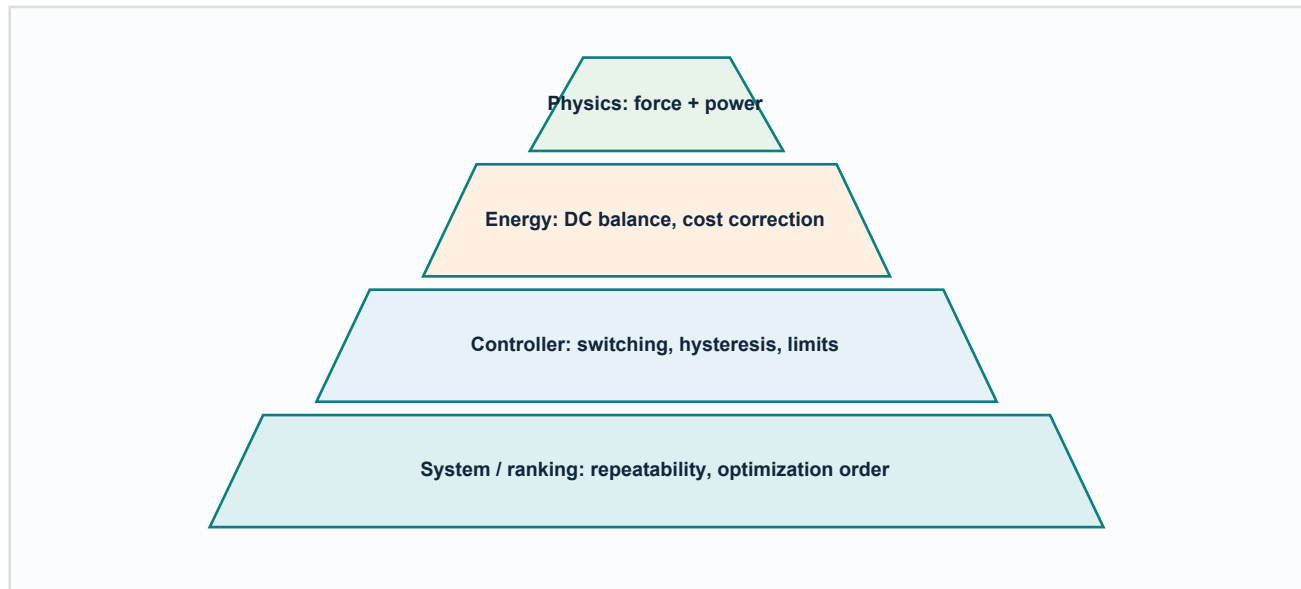


Figure 14.1. Verification pyramid used by the automated suite

The automated suite combines analytical comparisons, behavioral tests, traction-isolation tests, constant genset-power checks, saturation tests, data-validation failures, conservation checks, repeatability, terminal-cost checks, charge-sustaining logic, and optimizer ranking. All twenty-one scenarios currently pass.

## 14.2 Test matrix

**Table 14.1. Automated MATLAB test suite**

| Test                       | Purpose                                                        | Result |
|----------------------------|----------------------------------------------------------------|--------|
| ConstantSpeedLevel         | Constant-speed wheel power matches hand calculation            | PASS   |
| StationaryAuxiliary        | Auxiliary load consumes battery energy at zero speed           | PASS   |
| LevelAcceleration          | Acceleration increases wheel demand                            | PASS   |
| PositiveGrade              | Positive grade increases wheel power                           | PASS   |
| DownhillRegeneration       | Negative grade produces regenerative DC power                  | PASS   |
| BatterySwitching           | Low active pack deterministically switches to standby pack     | PASS   |
| StandbyGensetCharge        | Genset charges only the standby battery                        | PASS   |
| GensetTractionIsolation    | Genset output never offsets traction-bus demand                | PASS   |
| GensetHysteresis           | Genset starts for low standby SOE without chattering           | PASS   |
| BatteryUpperLimit          | Battery energy never exceeds upper limit                       | PASS   |
| BatteryLowerLimit          | Battery energy never falls below lower limit                   | PASS   |
| MotorSaturation            | Motor saturation reports unmet wheel power                     | PASS   |
| GensetBestEfficiencyPoint  | Genset runs at one constant optimum power                      | PASS   |
| RouteValidationFailure     | Duplicate route time is rejected                               | PASS   |
| IncompatibleSelection      | Compatibility filter records rejection reason                  | PASS   |
| EnergyConservation         | Cumulative DC residual remains within tolerance                | PASS   |
| Repeatability              | Identical inputs produce identical scalar KPIs                 | PASS   |
| ZeroSpeedStability         | Zero-speed calculation remains finite                          | PASS   |
| TerminalCostCorrection     | Grid-equivalent terminal correction matches hand calculation   | PASS   |
| ChargeSustainingComparison | Terminal combined-SOE tolerance is enforced                    | PASS   |
| OptimizationRanking        | Feasible configurations are sorted by total cost per kilometre | PASS   |

## 14.3 Independent hand check

For the current 19,000 kg default selection, the constant-speed level-road test compares the model wheel power against an independent analytical expression. Both calculations give 20.699 kW at the test condition. The test also confirms that the selected total mass enters the dynamics directly.

### Passing tests do not validate synthetic data

Tests show that the implementation behaves consistently with its equations and stated limits. They do not prove that catalog efficiencies, BSFC values, component ratings, or route assumptions represent a production vehicle.

# 15 Worked Example: Default VECTO Urban Case

## 15.1 Case definition

The default case uses the VECTO Urban route, two BAT-12 packs, two MOT-12 rear hub motors, GEN-05, a 19,000 kg total vehicle mass, 85% initial Battery 1 SOE, 75% initial Battery 2 SOE, 1.65 EUR/L diesel, and 0.18 EUR/kWh electricity.

## 15.2 Signal interpretation

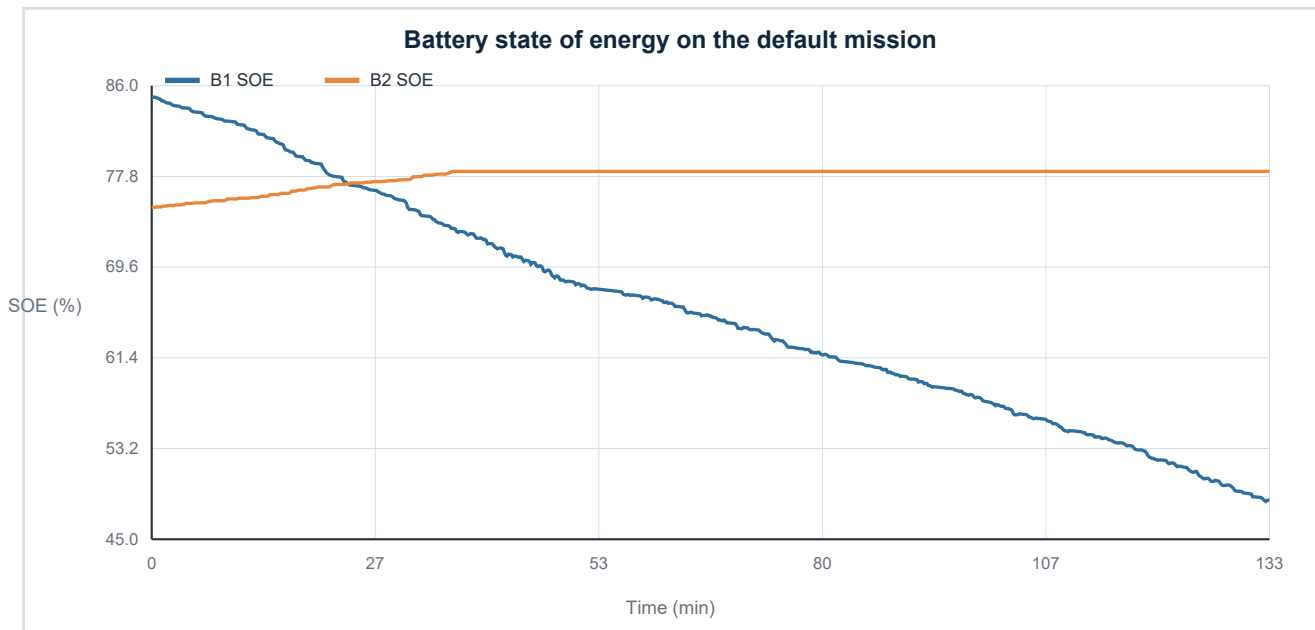


Figure 15.1. Default dual-battery SOE histories and active-pack behavior

Battery 1 supplies most traction demand and falls from 85% to approximately 48.5%. Battery 2 accepts regenerative energy and rises from 75% to approximately 78.2%. The genset does not start because the standby pack remains above the fixed 30% start condition.

## 15.3 Summary metrics

Table 15.1. Default-case results

| Metric                 | Value           | Interpretation                                       |
|------------------------|-----------------|------------------------------------------------------|
| Route distance         | 39.787 km       | distance integrated from the prepared 1 s trace      |
| Grid-equivalent energy | 75.957 kWh      | terminal pack deficits divided by grid efficiency    |
| Electrical intensity   | 1.9091 kWh/km   | equivalent replenishment basis                       |
| Operating cost         | 0.3436 EUR/km   | fuel plus grid electricity divided by distance       |
| Regenerated energy     | 25.584 kWh      | negative motor DC energy accepted before pack losses |
| Auxiliary energy       | 20.815 kWh      | base and HVAC electrical demand                      |
| Unmet traction energy  | 0.000000 kWh    | zero: case is feasible under the project criterion   |
| Energy-balance error   | 0.000000000 kWh | zero within numerical reporting precision            |

**Worked-example conclusion**

The case is internally feasible and conservative under equivalent replenishment. It is not charge sustaining because combined terminal SOE is below the initial value, and it is not a real-vehicle performance claim.



# 16 Long-Route and Mass Feasibility Studies

## 16.1 Mass sweep

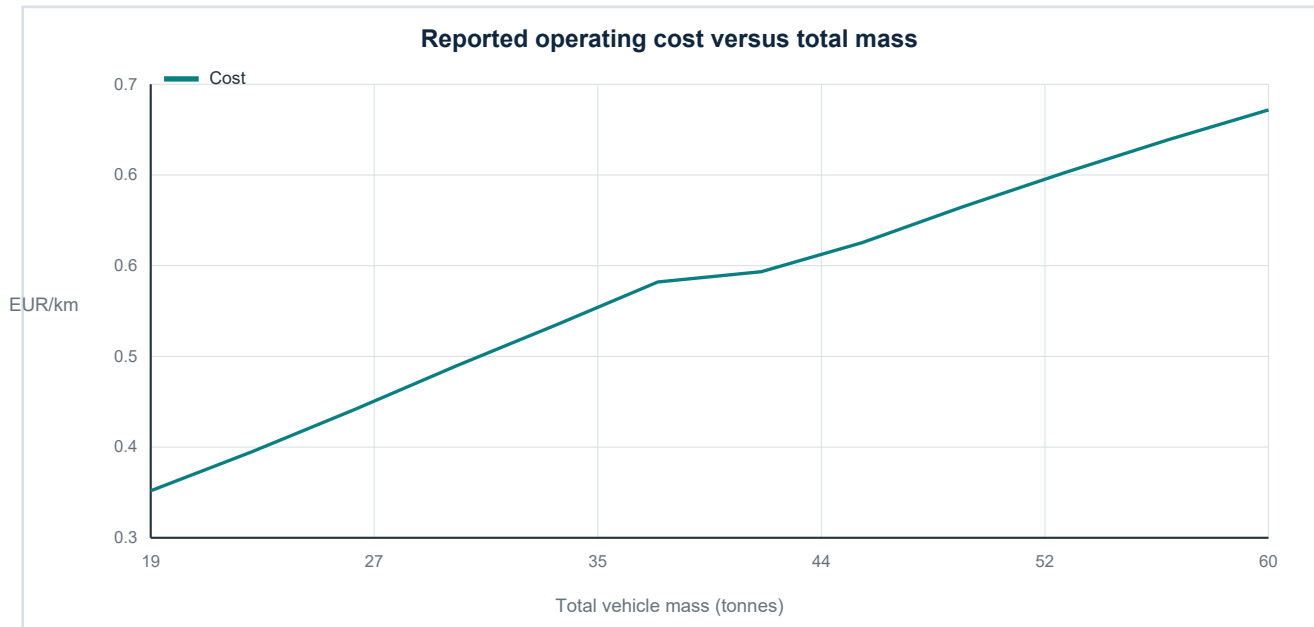


Figure 16.1. Cost trend must be read together with unmet traction energy

At 19 tonnes the default route has zero unmet traction energy. Small shortfalls begin near the next mass variant and reach 21.424 kWh at 60 tonnes. A cost trend alone therefore mixes feasible and infeasible cases; the feasibility flag is the governing result.

## 16.2 European long-route study

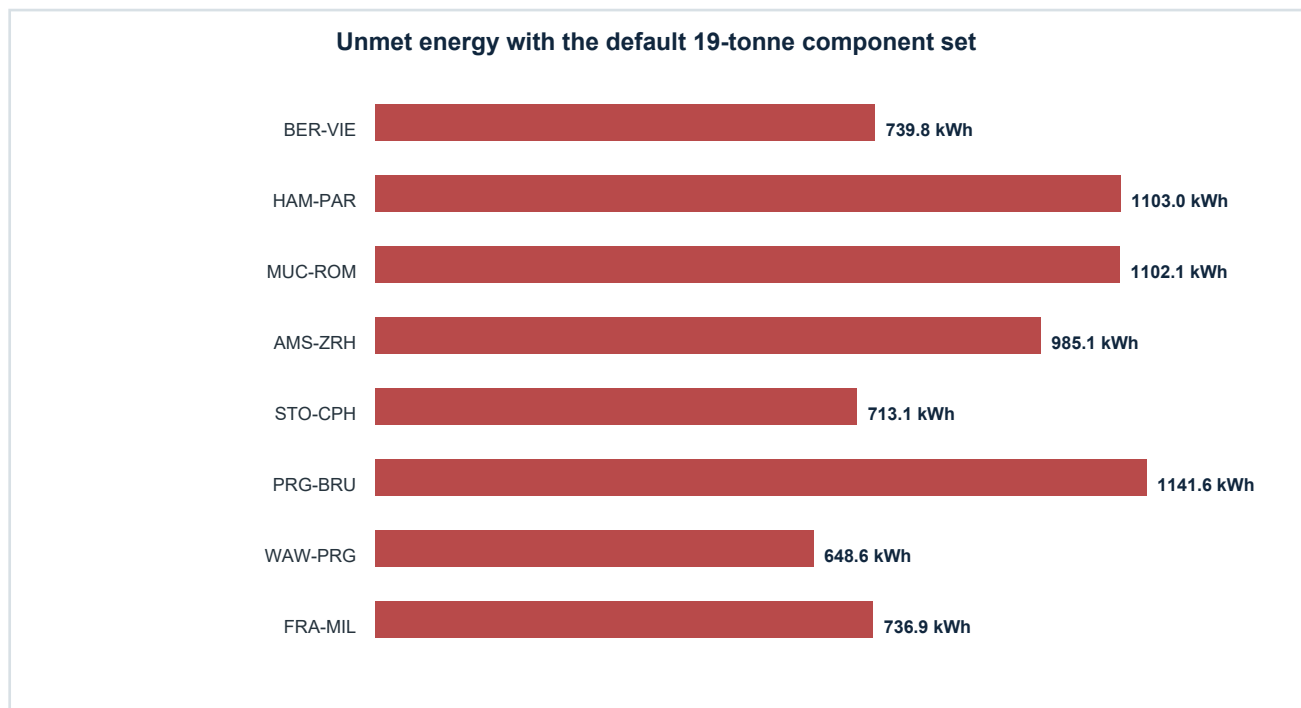


Figure 16.2. All long-route default cases are infeasible despite apparently low reported costs

The long missions integrate correctly, but the current default architecture cannot supply the complete requested traction/DC energy after stored energy is depleted. Unmet energy ranges from roughly 649 to 1,142 kWh. The low cost values in these runs are therefore not design candidates; they are examples of why feasibility must precede economics.

**Table 16.1. Default-component long-route outcomes**

| Route       | Simulated km | Fuel (L) | Unmet (kWh) | Feasible? |
|-------------|--------------|----------|-------------|-----------|
| EUR-BER-VIE | 677.4        | 8.0      | 739.8       | No        |
| EUR-HAM-PAR | 902.2        | 0.0      | 1103.0      | No        |
| EUR-MUC-ROM | 915.0        | 0.0      | 1102.1      | No        |
| EUR-AMS-ZRH | 820.9        | 0.0      | 985.1       | No        |
| EUR-STO-CPH | 656.3        | 0.0      | 713.1       | No        |
| EUR-PRG-BRU | 906.8        | 0.0      | 1141.6      | No        |
| EUR-WAW-PRG | 638.1        | 34.7     | 648.6       | No        |
| EUR-FRA-MIL | 664.8        | 0.0      | 736.9       | No        |

### 16.3 How to make a long-route study meaningful

- 1 Increase motor and active-pack capability, add sufficient battery energy, or schedule external charging; do not count genset output as traction power.
- 2 Introduce realistic elevation and road-grade data before comparing Alpine corridors.
- 3 Calibrate speed, dwell, passenger load, HVAC, traffic, and rest-stop assumptions with measured or operator data.
- 4 Run bounded optimization with feasibility constraints, then compare equivalent replenishment and charge-sustaining results.
- 5 Repeat under temperature, mass, headwind, and component-degradation scenarios.

# 17 Limitations, Responsible Use, and Extensions

## 17.1 Current model boundary

**Table 17.1. Fidelity gaps and extension paths**

| Missing fidelity                        | Why it matters                                                             | Recommended extension                                       |
|-----------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|
| Battery voltage/current and temperature | Power and efficiency limits change with electrical and thermal state       | Equivalent-circuit and thermal network with calibrated maps |
| Ageing                                  | Usable energy, resistance, and cost evolve with throughput and temperature | Cycle/calendar degradation and life-cycle cost              |
| Closed-loop driver                      | Backward demand may exceed trackable acceleration or actuator limits       | Driver controller and forward vehicle dynamics              |
| Road elevation for long routes          | Mountain energy and regeneration are understated                           | Map-matched elevation with smoothing and grade validation   |
| Engine transients and emissions         | Fuel and pollutants depend on warm-up and aftertreatment                   | Dynamic engine/aftertreatment model and measured maps       |
| Wheel slip and lateral dynamics         | Traction limits and stability are not represented                          | Tyre-road force limits and vehicle dynamics                 |
| Supplier data                           | Synthetic catalogs cannot support procurement                              | Version-controlled, validated manufacturer maps             |

## 17.2 Recommended development roadmap

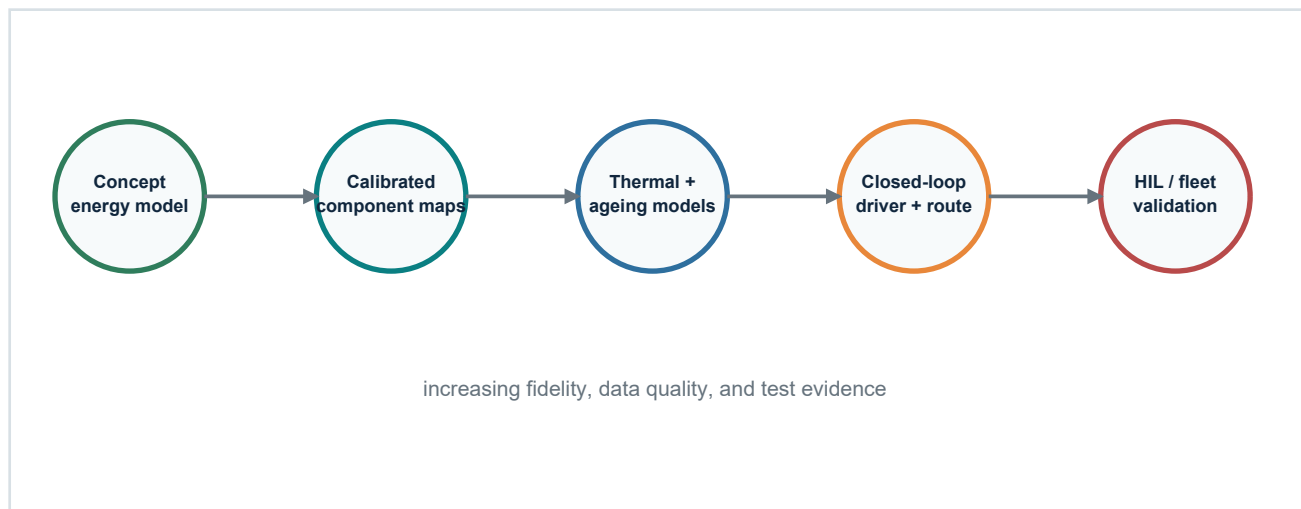


Figure 17.1. A disciplined progression from concept model to validated engineering platform

## 17.3 Graduate exercises

- 1 Derive an analytical constant-speed energy-per-kilometre expression and compare it with the kernel over several masses and headwinds.
- 2 Replace scalar motor efficiency with a two-dimensional torque-speed map and quantify the change in route energy.
- 3 Add elevation to one OSM/OSRM route, validate grade smoothing, and compare energy with the zero-grade baseline.
- 4 Design an alternative supervisor that allows controlled parallel discharge and define new safety/feasibility tests.
- 5 Formulate a multi-objective search for cost, fuel, battery throughput, and component mass; produce a Pareto front.
- 6 Create a charge-sustaining urban case by sizing battery energy and the fixed-point standby charger; keep the 30% role threshold unchanged.
- 7 Add a simple battery thermal state and demonstrate a cold-weather power-derating scenario.
- 8 Construct a requirements-to-test traceability matrix for every limit and KPI in the model.

### Final engineering lesson

A useful model is not only a set of equations. It is an agreement among data definitions, physical assumptions, interfaces, numerical methods, controller logic, evidence, and interpretation. When any one of those changes, the others must be reviewed.

# Appendix A Quick-Start Laboratory

---

Use the following sequence for a first supervised laboratory session.

- 1 Open MATLAB R2025a and change to the project folder.
- 2 Run `startup_hybrid_bus` and confirm that the project loads without startup issues.
- 3 Run `Results = run_hybrid_bus_simulation`; inspect `Results.Summary` and `Results.Validation`.
- 4 Open `HybridBus_BackwardModel.slx` and trace signals from route input to energy accounting.
- 5 Launch `app = HybridBusApp`; repeat the default case and inspect all plot tabs.
- 6 Change only one factor, such as mass or route, and explain every KPI change before running a second factor.
- 7 Run `TestResults = run_all_hybrid_bus_tests`; confirm 21 PASS results.

```
db = load_hybrid_bus_database("HybridBus_ComponentDatabase.xlsx");
report = validate_hybrid_bus_database(db);
assert(report.IsValid, strjoin(report.Errors,newline));

overrides = struct('SelectedRoute',"EUR-BER-VIE", ...
                  'SelectedMass',"MASS-01");
Results = run_hybrid_bus_simulation( ...
    "HybridBus_ComponentDatabase.xlsx",overrides,'SaveResults',false);
```

# Appendix B Signal and Parameter Reference

**Table B.1. Principal logged signals**

| Group       | Signal                                | Unit       | Meaning                                            |
|-------------|---------------------------------------|------------|----------------------------------------------------|
| Vehicle     | Speed_m_s / Acceleration_m_s2         | m/s, m/s2  | prescribed speed and filtered acceleration         |
| Wheel       | Demand_kW / Delivered_kW              | kW         | required and limit-delivered wheel power           |
| Wheel       | UnmetTraction_kW / UnmetRegen_kW      | kW         | mechanical shortfalls                              |
| Motors      | ElectricalPower_kW / MotorSpeed_rpm   | kW, rpm    | DC demand/regeneration and motor speed             |
| Auxiliary   | Power_kW                              | kW         | base, HVAC, and route-multiplied load              |
| Battery 1/2 | Power_kW / Energy_kWh / SOE           | kW, kWh, 1 | pack source/sink and stored state                  |
| Genset      | ElectricalPower_kW / FuelRate_L_s     | kW, L/s    | constant standby-charger power and fuel flow       |
| Genset      | ChargeDestinationBattery              | integer    | standby pack receiving genset power; zero when off |
| Controller  | ActiveBattery / StandbyBattery / Mode | integer    | supervisory role and discrete mode                 |
| Energy      | BalanceResidual_kW                    | kW         | power-conservation diagnostic                      |
| Energy      | UnmetDCPower_kW / RejectedCharge_kW   | kW         | source/sink saturation accounting                  |

**Table B.2. Principal parameter groups**

| Area       | Principal parameters                                               | Workbook source                             |
|------------|--------------------------------------------------------------------|---------------------------------------------|
| Vehicle    | mass, gravity, Cd, frontal area, rolling resistance                | Mass / Tyre / Vehicle                       |
| Route      | time, speed, grade, dwell, auxiliary multiplier                    | Route sheets                                |
| Hub drive  | wheel radius, ratio, efficiencies, torque/power/speed              | Tyre / Final Drive / Motor                  |
| Battery    | usable energy, SOE bounds, efficiencies, power, regen              | Battery + Dashboard                         |
| Genset     | optimum/max power, minimum times, BSFC, efficiency                 | Genset + maps                               |
| Supervisor | fixed 30% role threshold, standby upper-SOE target, regen redirect | Architecture rule + control calibration     |
| Economics  | fuel tank, prices, grid efficiency, terminal method                | Dashboard / Vehicle / Prices / Optimization |

# References Sources and Project Files

---

- [1] HybridBusProject source tree, workbook, MATLAB functions, Simulink model, and generated results. Database version 1.4.0, revised 15 August 2026.
- [2] European Commission, Vehicle Energy Consumption Calculation Tool (VECTO), [https://climate.ec.europa.eu/eu-action/transport-decarbonisation/road-transport/vehicle-energy-consumption-calculation-tool-vecto\\_en](https://climate.ec.europa.eu/eu-action/transport-decarbonisation/road-transport/vehicle-energy-consumption-calculation-tool-vecto_en).
- [3] European Commission VECTO source repository, declaration mission cycles, commit d16ba783c1d8af0ea68797f5d7ed6cf01877d402, <https://code.europa.eu/vecto/vecto>.
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- [5] OpenStreetMap contributors, Copyright and Open Database License attribution, <https://www.openstreetmap.org/copyright>.
- [6] European Commission Mobility and Transport, Driving time and rest periods, including a 45-minute break after 4.5 hours of driving, [https://transport.ec.europa.eu/transport-modes/road/social-provisions/driving-time-and-rest-periods\\_en](https://transport.ec.europa.eu/transport-modes/road/social-provisions/driving-time-and-rest-periods_en).
- [7] Project documentation: System\_Architecture.md, Mode\_Transition\_Table.md, Parameter\_Dictionary.md, Signal\_Dictionary.md, Assumptions\_and\_Limitations.md, and Test\_Report.md.